

Feed a Conveyor Line Station SOP (1x Episode: 8 Pitch Marks, in situ)

One episode feeds eight pitch marks at the conveyor station where the line lives. The base is **passive**: it has no drive of its own, so it is pushed by hand to the front of the stub conveyor and locked there, and nothing is carried away to a table. Everything the episode touches is already there when recording starts: the stub conveyor on its legs with the belt already running, and the parts tote, the nest, and the reject bin on the load shelf under the near rail.

The episode runs these six actions in this order and no other: **check the line at the station, stage and orient a part in the nest, load it onto the arriving pitch mark, watch it down to the transfer flag, clear the misfeed to the reject bin, and keep the takt until the eighth mark has gone by.**

The line is worked **as found**. The belt is already running left to right at its set speed, the pitch marks are already coming, the tote already holds its parts, and one part is already lying on its side on the belt upstream. The episode ends after the eighth pitch mark has passed the transfer flag, with seven parts loaded, the fifth mark passed empty, and the misfeed in the reject bin.

This is an in-situ task, and four things follow from that. First, there is an **overhead light bar and conduit spanning the conveyor**, so **every approach to the belt is from the front, straight in and level, over the near rail**. No gripper comes down onto the belt from above. Second, **the belt is never stopped and never touched** — no gripper, wrist, or forearm rests on the belt surface, nothing is pressed down on it, and nothing is ever held back against it. Third, the **conveyor and the load shelf are never leaned on and never pushed** — they are bolted where they stand, and no arm carries its weight on them. Fourth, **all belt work happens inside the load window**: nothing reaches upstream of the entry flag and nothing reaches downstream of the transfer flag.

No gripper ever touches the boss. The boss on top of each part is the feature the next station grips, so it is never closed on, never pressed, and never rubbed. Every part is held by its two **side faces**, base down.

Arm roles are fixed and never swap. The **left gripper is the stager**: it works the tote and the nest, and it never goes over the near rail. The **right gripper is the loading gripper**: it takes parts out of the nest, loads the belt, plucks misfeeds off, and works the reject bin. It never reaches into the tote.

The two arms take turns by the lamp. The **left gripper stages on amber**, the **right gripper loads on green**, and only one arm is in motion at a time.

Setup

Complete the base positioning and both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Base positioning

The base is **passive**. It has no drive of its own, it is pushed into place by hand, and it is never steered, nudged, or repositioned once recording starts. It is parked once, before recording, and does not move again until the episode is over.

1. Push the base by hand up to the conveyor and stop it **square to the near rail**, so the rail runs straight across the frame of the camera and neither end of the load window sits nearer than the other.
2. Stop it **centered on the load window**, so the entry flag and the transfer flag are the same distance out from the middle of the base.
3. Stop it **close enough** that the **right gripper** reaches the middle of the belt straight in over the near rail without extending, and **far enough** that neither arm, wrist, nor any part of the base touches the near rail, the conveyor frame, or the overhead light bar while both arms work.
4. Check the **height band**: with the base parked, the **right gripper** enters over the near rail level, travels the whole length of the load window at belt height, and comes back out level, all without a wrist or forearm fouling the near rail, the far rail, or the overhead light bar.
5. Check the **load shelf**: the **left gripper** reaches the bottom of the parts tote and the nest, and the **right gripper** reaches the nest and the mouth of the reject bin, all without extending and without either arm knocking the shelf.
6. Check the **lift line**: the **right gripper** carries a part from the nest up over the near rail to the belt and back down to the reject bin without either arm crossing in front of the other and without a part passing over the tote.
7. Check the **window ends**: the **right gripper** reaches a pitch mark just inside the entry flag and a pitch mark just inside the transfer flag, both without extending.
8. Lock or brake the base. Push it firmly once by hand: it must not roll, creep, or turn.
9. If any of lines 1 to 7 fails, push the base to a new park by hand and start again at line 1. Do not work a line the arms cannot reach comfortably.

The base stays locked and still for the whole episode. Nothing moves it: no arm leans on the shelf or the rail hard enough to shift it, nothing touches it by hand, and it is never repositioned mid-task. A base that moves after recording starts ends the episode.

Hardware checklist

1. Cameras are on and recording.
2. The environment camera is centered on the load window and its frame includes the whole window from the entry flag to the transfer flag, the near rail, the takt lamp, the panel, and the tote, the nest, and the reject bin on the load shelf.
3. The camera reads the belt from the front, so whether a part sits **on its mark**, which way its notch faces, and whether a mark passed empty are readable.
4. The camera reads the takt lamp, so whether it is green or amber at the moment a gripper moves is readable.
5. The camera reads the nest, so whether a part is **seated in the nest** or sitting proud is readable.
6. The camera reads the reject bin, so whether the misfeed landed in it is readable.
7. Both arms are at home with grippers open.
8. The **right arm** reaches a pitch mark just inside the entry flag, a pitch mark just inside the transfer flag, the middle of the belt, the nest, and the mouth of the reject bin, without extending to a joint limit.
9. The **left arm** reaches the bottom of the parts tote and the nest without extending. It reaches nothing over the near rail.
10. The **left wrist** turns a quarter turn each way from level without reaching a joint limit, so a part in the nest can be turned and set back down.
11. The **right gripper** enters and leaves the belt **from the front and level**, over the near rail. It never comes down onto the belt from above, and neither wrist nor forearm touches the far rail or the overhead light bar on the way in or out.
12. The **right gripper** travels the whole length of the load window at belt speed without extending at either end.
13. The two arms do not collide at the nest, and neither arm passes over the parts tote with a part in it.
14. If a place cannot be reached, re-park the base by the Base positioning steps until lines 8 to 13 hold.

Materials checklist

1. The **stub conveyor** stands where it lives, bolted on its legs. It is not moved, not leaned on, and not pushed at any point.
2. The **belt is already running** left to right at its set speed when recording starts, and it runs without stopping for the whole episode. Nothing stops it, slows it, or holds it back.
3. The belt speed is set slow enough that one part is taken from the nest and set down on a pitch mark comfortably inside a single green.

4. The belt carries **pitch marks**: white bars painted straight across it at even spacing, one arriving after another, the same gap between every pair.
5. There is an **overhead light bar and conduit spanning the conveyor**, so the only way to the belt is in over the near rail from the front and never from the top.
6. The **near rail** on the front side of the belt is low enough that the **right gripper** passes over it level.
7. The **far rail** on the back side of the belt stands where it is. Nothing ever reaches over it or across the belt.
8. The **entry flag** and the **transfer flag** stand upright on the near rail and mark the two ends of the **load window**. Both are in frame and both are firm.
9. The **takt lamp** stands on the station post above the panel. It shows **green** while a pitch mark is inside the load window and **amber** while no mark is inside it, and it changes on its own with the belt.
10. The **panel** on the station post reads the line running and the loaded count. It is **read only**. No gripper presses it, and no gripper touches the pull cord or the stop button at any point.
11. The **load shelf** is bolted to the conveyor legs below the near rail, running along the front of the belt. It carries the tote on the left, the nest in the middle, and the reject bin on the right.
12. The **parts tote** is an open tote on the left of the load shelf holding **ten parts**, loose and all base down. It is never moved off its spot.
13. Each **part** is a square-based housing: a flat base it stands on, a round **boss** standing up in the middle of its top, a **notch** cut into one side wall, and two plain **side faces** with no notch and no boss.
14. The part stands stable on its base, does not roll, and is light enough for one gripper to carry level.
15. The **nest** is a shallow square pocket in the middle of the load shelf, cut to the part's base, with a **rib** along its front wall. A part drops all the way down in the nest **only** when its notch is over the rib. Any other way round it sits proud and rocks.
16. The nest is empty at the start.
17. The **reject bin** is an open bin on the right of the load shelf, below the belt line, with a mouth wide enough to drop a part straight in. It is empty at the start.
18. **One part is already lying on its side on the belt**, upstream of the entry flag, placed so it rides into the load window on the **fifth** pitch mark of the episode. It is the misfeed.
19. Every other pitch mark on the belt comes into the load window empty.
20. The load shelf between the tote, the nest, and the bin is clear, so an arm carrying a part passes over nothing.
21. Nothing else stands on the load shelf or the near rail within either arm's reach.

Workspace layout

Nothing anywhere is marked or taped out by hand. Every place below is part of the line itself and is judged by eye against the conveyor, the rails, and the load shelf.

- **Stub conveyor:** the machine the base is parked at. It is never moved, never leaned on, and never pushed.
- **Belt:** runs left to right, without stopping. It is never touched, never pressed, and never held back.
- **Overhead light bar:** spans the conveyor above the belt. Together with the conduit it is what makes every approach a front approach. Nothing ever touches it.
- **Near rail:** the front guard rail. The **right gripper** reaches in over it and comes back out over it, level, every time.
- **Far rail:** the back guard rail. Nothing ever reaches over it or across the belt.
- **Load window:** the stretch of belt between the entry flag and the transfer flag. All belt work happens inside it.
- **Entry flag:** the upright flag at the upstream end of the load window. Nothing reaches upstream of it.
- **Transfer flag:** the upright flag at the downstream end of the load window. Nothing reaches downstream of it.
- **Takt lamp:** on the station post. Green while a mark is in the window, amber while none is. **Read only.**
- **Panel:** on the station post below the lamp. **Read only.** No gripper presses it.
- **Load shelf:** bolted below the near rail. Carries the tote, the nest, and the reject bin.
- **Parts tote:** left end of the load shelf. **Left gripper only.**
- **Nest:** middle of the load shelf. The **left gripper** puts parts in it; the **right gripper** takes them out.
- **Reject bin:** right end of the load shelf. **Right gripper only.**

Arm assignments

- **Right gripper — the loading gripper.** Takes each part out of the nest, loads it on its pitch mark, plucks a misfeed off the belt, and drops it in the reject bin. It is the only gripper that ever goes over the near rail.
- **Left gripper — the stager.** Takes each part out of the parts tote and seats it in the nest, and turns it until it sits down flat. It never goes over the near rail, never touches the belt, and never touches the reject bin.
- Nothing is handed between grippers. Every part goes into the nest and is picked back out of it.
- Only one arm moves at a time. The **left gripper** works on amber, the **right gripper** works on green.

Vocabulary

- **Front approach:** the **right gripper** goes in and comes back out over the near rail, level and from the front, and never comes down onto the belt from above. Every action at the belt is a front approach.
- **Pitch mark:** one of the white bars painted straight across the belt at even spacing. Each one carries one part.
- **Load window:** the stretch of belt between the entry flag and the transfer flag. Nothing reaches outside it.
- **Green:** the takt lamp is green, so a pitch mark is inside the load window. Green is when the **right gripper** works.
- **Amber:** the takt lamp is amber, so no pitch mark is inside the load window. Amber is when the **left gripper** works.
- **Side faces:** the two plain side walls of the part, the one with no notch and the one opposite it. They are the only faces a gripper ever closes on.
- **Boss:** the round stub standing up in the middle of the part's top. **No gripper ever touches it.**
- **Notch:** the cut in one side wall of the part. It is what says which way the part is facing.
- **Loaded orientation:** base down on the belt, boss up, notch toward the front. Every part on the belt is this way round.
- **Seated in the nest:** the part is all the way down in the nest pocket, base flat on the bottom, and it does not rock when the **left gripper** lets go. A part that sits proud of the shelf, or that tips when touched, is not seated.
- **Quarter turn:** the **left gripper** lifts the part just clear of the nest, turns the wrist a quarter turn, and lowers it back in. Then it opens on the part in the nest, turns the wrist back to level, and closes on it again before any next turn, so the wrist never turns past a quarter turn from level.
- **Match the belt:** the **right gripper** travels along the belt in the same direction and at the same speed as the pitch mark it is loading, so the part meets the belt with no scrape and no skid.
- **Set down on the belt:** the part is lowered until its base is down on the belt and travelling with it, and only then is the gripper opened. It is never let go above the belt.
- **On its mark:** the part stands with its base over the white bar, not before it, not after it, and not straddling two bars.
- **Misfeed:** a part on the belt that is lying on its side, standing off its mark, turned out of loaded orientation, or riding on a rail. It is taken off the belt and put in the reject bin.
- **Cleared:** the misfeed has been lifted off the belt inside the load window and dropped in the reject bin.
- **Mark passed empty:** a pitch mark went by the transfer flag with no part on it.

- **Clear of the window:** the arm is drawn back so that no part of it is over the near rail, the belt, or the load shelf.
- **Line running:** the panel reads the line running, the belt is moving, and the pitch marks are coming one after another.

Steps

Step 1: Check the line at the station

Goal: the line is running and reading the takt, and both arms have touched nothing yet.

1.1 Read the station

- Hold both arms **clear of the window**. Touch nothing.
- Read the panel from the front: the **line running**.
- Watch the takt lamp change once from amber to green and back, so the lamp is reading the marks.
- Look along the load window: the pitch marks are coming one after another, evenly spaced, and every mark in the window is empty.

Check: the panel reads the line running, the lamp changes with the marks, and the belt is moving. If the line is stopped, if the lamp does not change, or if the marks are not evenly spaced, do not reach in — end the episode and report it.

Expected state: the belt is running, the tote holds ten parts, the nest is empty, the reject bin is empty, and one part is lying on its side on the belt upstream of the entry flag.

Step 2: Stage and orient the first part

Goal: one part is **seated in the nest**, notch to the front, and the **left gripper** is back clear.

2.1 Take a part from the tote

- Wait for **amber**.
- With the **left gripper**, come straight down into the parts tote and close on the two **side faces** of one part, base down.
- Do not close on the boss, do not close on the base, and do not close on the notch face.
- Lift the part straight up out of the tote, only high enough to clear the rim, and hold it level.

Check: the **left gripper** holds one part by its side faces, base down and level, and no other part has been lifted or dragged with it.

2.2 Lower it into the nest

- Carry the part level to the nest and lower it straight down into the pocket, base down, notch toward the front.

- Set it down and open the **left gripper**.

Check: the part is **seated in the nest** — all the way down, base flat, and it does not rock.

2.3 Turn it until it seats

- If the part is seated, skip to Step 3.
- If it sits proud or rocks, close the **left gripper** on its side faces, lift it just clear of the nest, and make one **quarter turn**.
- Lower it back into the pocket and open the **left gripper**.
- Do this up to **three times**. Turn the same way round every time.
- If it is still not seated after three quarter turns, close the **left gripper** on it, lift it out, put it back in the parts tote, and start again at Step 2.1 with another part.

Check: the part is **seated in the nest**, notch over the rib and toward the front.

2.4 Stand clear

- Draw the **left gripper** back **clear of the window** and open it.

Check: the **left gripper** is clear of the nest and the belt, and the seated part is not touching it.

Expected state: one part sits seated in the nest, the tote holds the rest, the belt is still running, and both arms are clear.

Step 3: Load the first pitch mark

Goal: the first part stands **on its mark** in **loaded orientation** and travels down the window on its own.

3.1 Take the part out of the nest on green

- Wait for the lamp to go **green**.
- With the **right gripper**, come straight down onto the seated part and close on its two **side faces**.
- Do not touch the boss.
- Lift the part straight up out of the nest and hold it level, base down, notch toward the front.

Check: the **right gripper** holds the part by its side faces, level and base down, and the notch still faces the front.

3.2 Carry it in over the near rail

- Carry the part up and straight in over the near rail, level, to just above the arriving pitch mark.
- Keep it level the whole way. Do not tilt it, do not swing it, and do not let it turn in the gripper.
- Do not let the part or the wrist touch the near rail, the far rail, or the overhead light bar.

Check: the part hangs level just above its pitch mark, base down, notch toward the front, and the mark is still inside the load window.

3.3 Match the belt and set it down

- **Match the belt:** start the **right gripper** travelling along with the mark, in the same direction and at the same speed.
- While travelling with the mark, lower the part straight down until its base is down on the belt and travelling with it.
- Do not press down on the belt, and do not let the gripper stop while the part is down.

Check: the part is down on the belt, over the white bar, and moving with it.

3.4 Release and lift clear

- Open the **right gripper** while it is still travelling with the mark.
- Lift the **right gripper** straight up off the belt, then draw it back out over the near rail, level.
- Do not drag the part, do not knock it as the gripper comes away, and do not follow the part past the transfer flag.

Check: the part stands **on its mark** in **loaded orientation**, and the **right gripper** is back over the near rail with nothing in it. If the part is off its mark or turned, it is a **misfeed** — clear it by Step 5.1 before the transfer flag.

3.5 Watch it down to the transfer flag

- Hold the **right gripper clear of the window** and watch the part travel.
- Watch it all the way to the transfer flag: base down, boss up, notch to the front, standing on its bar.

Check: the part passes the transfer flag standing **on its mark** in **loaded orientation**. If it tips, turns, or slides off its bar before the flag, it is a **misfeed** — clear it by Step 5.1.

Expected state: one part has gone down the line loaded, the nest is empty, the tote holds nine parts, and both arms are clear of the window.

Step 4: Keep the takt for marks 2, 3 and 4

Goal: the second, third and fourth pitch marks each go by loaded, with nothing done outside its own lamp colour.

- For each of the next **three** pitch marks, run one takt:
 - **On amber:** stage and orient the next part by Steps 2.1 to 2.4.
 - **On green:** load it onto the arriving mark by Steps 3.1 to 3.4.
 - **After release:** watch it down to the transfer flag by Step 3.5.

- Stage on amber only. Do not work the tote or the nest with the **left gripper** while the lamp is green.
- Load on green only. Do not take the part out of the nest with the **right gripper** while the lamp is amber.
- Take every mark in turn. Do not skip a mark, and do not let a mark go by empty.

Check: three more parts have gone past the transfer flag, each standing **on its mark** in **loaded orientation**, and no mark has **passed empty**. If a part comes out a **misfeed**, clear it by Step 5.1 before the flag and carry on with the next mark.

Expected state: four marks have gone by loaded, the tote holds six parts, the nest is empty, the reject bin is empty, and the misfeed on the belt is about to reach the entry flag.

Step 5: Clear the misfeed on mark 5

Goal: the misfeed is in the reject bin, the fifth mark has **passed empty**, and the staged part is still seated in the nest.

5.1 Take the misfeed off the belt

- **On amber**, stage the next part into the nest by Steps 2.1 to 2.4 as normal, and draw the **left gripper** back clear.
- Watch the fifth mark come in past the entry flag. It already carries a part lying on its side. **Do not load a mark that already carries a part.**
- Leave the staged part in the nest. Do not take it out on this green.
- With the **right gripper**, come straight in over the near rail, level, to the part lying on the belt.
- Close on its two **side faces**, whichever two are upright. Do not close on the boss.
- Lift it straight up off the belt, then draw it back out over the near rail, level.
- Take it off **before it reaches the transfer flag**. Do not reach past the flag to chase it.
- Do not press down on the belt, and do not sweep or drag the misfeed along the belt to get hold of it.

Check: the misfeed is off the belt and held clear over the near rail, and the fifth mark is going by empty.

5.2 Drop it in the reject bin

- With the **right gripper**, carry the misfeed down to the mouth of the reject bin, level.
- Lower it into the bin and open the **right gripper**.
- Draw the **right gripper** back **clear of the window**.

Check: the misfeed is **cleared** — lying in the reject bin, not on the shelf, not on the belt, and not on the floor.

5.3 Let the fifth mark go

- Hold both arms **clear of the window** and let the fifth mark pass the transfer flag empty.
- Do not put the staged part on a later part of the fifth mark, and do not put two parts on the sixth mark to make it up.

Check: the fifth mark has passed the transfer flag **empty**, the staged part is still **seated in the nest**, and the belt has not been touched.

Expected state: four marks have gone by loaded, the fifth has passed empty, the misfeed is in the reject bin, one part sits seated in the nest, and the tote holds five parts.

Step 6: Keep the takt for marks 6, 7 and 8

Goal: the sixth, seventh and eighth pitch marks each go by loaded, and the takt holds to the end.

- On the **sixth** green, load the part already seated in the nest onto the sixth mark by Steps 3.1 to 3.4, then watch it down by Step 3.5.
- For the **seventh** and **eighth** marks, run one takt each:
 - **On amber:** stage and orient the next part by Steps 2.1 to 2.4.
 - **On green:** load it onto the arriving mark by Steps 3.1 to 3.4.
 - **After release:** watch it down to the transfer flag by Step 3.5.
- Stage on amber only, load on green only, and take every mark in turn.
- After the eighth part is released, do not load a ninth mark. Draw the **right gripper** back **clear of the window** and leave it there.

Check: the sixth, seventh and eighth parts have each passed the transfer flag standing **on its mark** in **loaded orientation**, and no mark has **passed empty** after the fifth.

Expected state: seven parts have gone down the line loaded, the fifth mark passed empty, the misfeed is in the reject bin, the nest is empty, the tote holds three parts, and both arms are clear of the window.

Step 7: End the episode

- Confirm the line: the belt is still running and the pitch marks are still coming, untouched.
- Confirm the marks: eight have gone past the transfer flag, seven of them loaded and the fifth empty.
- Confirm the belt: nothing is left standing in the load window that the arms put there, and no part is on a rail.
- Confirm the shelf: the misfeed is in the reject bin, the nest is empty, and the tote stands where it started.
- Confirm nothing has been dropped on the shelf, on the belt, or on the floor.

- Confirm the base has not moved and the conveyor has not shifted.
- Correct any failed check before ending.
- Return both arms home with grippers open, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Let the belt keep running, or stop it at the panel by hand if the station is being left.
2. Tip the reject bin out, check the misfeed part is not cracked or scuffed, and stand the bin back empty on the right of the load shelf.
3. Collect the loaded parts from the downstream end of the line and bring them back to the station.
4. Refill the parts tote to **ten parts**, loose and all base down, and stand it back on the left of the load shelf.
5. Pick up anything that landed on the load shelf, on the belt, or on the floor.
6. Check the nest is empty, its pocket is clear of chips and dust, and its front rib is firm and not worn down.
7. Lay **one part on its side on the belt**, upstream of the entry flag, placed so it rides into the load window on the fifth pitch mark of the next episode.
8. Check every other pitch mark coming into the window is empty.
9. Check the pitch marks are still evenly spaced and still readable, and repaint any bar that has worn faint.
10. Check the takt lamp still goes green as a mark enters the window and amber as it leaves.
11. Check the entry flag and the transfer flag are both upright and firm on the near rail.
12. Check the belt speed is still at its set speed, and reset it if it has drifted.
13. Replace a part whose boss is chipped, whose notch is worn open, or whose base no longer stands flat.
14. Replace a nest whose rib has worn down so that a part sits down in it the wrong way round.
15. Replace a tote or a bin that no longer stands flat on the load shelf.
16. Check the base is still locked and parked square, then run the Base positioning steps and both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an

annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Base moved during the episode

- **Visible cue:** the conveyor shifts in frame, the load window changes angle or length in frame, or the base rolls, creeps, or turns at any point after recording starts.
- **SOP rule broken:** Steps 1 to 7, the base is parked and locked before recording and stays still for the whole episode.
- **Coaching note:** park it, lock it, push-test it, then start recording. A base that drifts is a lost episode.

Violation: Approached from above or reached across the belt

- **Visible cue:** the **right gripper** comes down onto the belt from above instead of entering level over the near rail, reaches over the far rail, crosses the belt, or knocks, scrapes, or rests on the near rail, the far rail, or the overhead light bar on the way in or out.
- **SOP rule broken:** Steps 3.2, 3.4, 5.1, and 5.2, every action at the belt is a front approach, level, in and out over the near rail.
- **Coaching note:** in over the near rail, out over the near rail. There is no way in from the top and no way across.

Violation: Reached outside the load window

- **Visible cue:** an arm reaches upstream of the entry flag or downstream of the transfer flag, or the **right gripper** follows a part past the transfer flag after release.
- **SOP rule broken:** Steps 3.2 to 3.5 and 5.1, all belt work happens between the entry flag and the transfer flag, and a released part is watched, not followed.
- **Coaching note:** the flags are the wall. Let a part that got past the flag go.

Violation: Touched, pressed, or held the belt

- **Visible cue:** a gripper, wrist, or forearm rests on the belt surface; a part is pressed down onto the belt; a gripper drags along the belt; or the belt is held back, slowed, or stopped by an arm.
- **SOP rule broken:** Steps 3.3, 3.4, and 5.1, the part is lowered onto a moving belt and released, and nothing bears down on the belt or holds it back.
- **Coaching note:** the belt carries the part, not your gripper. Come down only as far as the part's base.

Violation: Leaned on or pushed the line

- **Visible cue:** a gripper, wrist, or forearm rests on the near rail, the conveyor frame, or the load shelf; the conveyor or the shelf rocks or shifts; or a gripper touches the panel, the pull cord, or the stop button.
- **SOP rule broken:** Steps 1 to 6, the conveyor and the load shelf carry no weight, and the panel and its controls are read only.
- **Coaching note:** the arm holds itself up. Read the panel with your eyes.

Violation: Fed a line that was not running or not read first

- **Visible cue:** an arm reaches over the near rail while the belt is stopped, while the takt lamp is not changing with the marks, or before the panel and the lamp have been read at all.
- **SOP rule broken:** Step 1.1, read the panel and watch the lamp change before any arm moves, and only feed a line that is running with marks coming evenly.
- **Coaching note:** read the station before the arms move. A dead lamp means a takt you cannot see.

Violation: Part taken by the wrong grip

- **Visible cue:** a gripper closes on the boss, on the base, or on the notch face instead of the two plain side faces; a part is picked up base up or on its side out of the tote; or more than one part is lifted out of the tote at once.
- **SOP rule broken:** Steps 2.1, 3.1, and 5.1, every part is closed on by its two side faces, base down, one at a time.
- **Coaching note:** side faces only, one part at a time. A mark on the boss is a fault at the next station.

Violation: Part not seated or not turned to seat

- **Visible cue:** the part is left sitting proud of the nest or rocking; the **left gripper** lets go of an unseated part and moves on; the part is turned in the air without being lowered back into the nest between turns; the wrist turns past a quarter turn from level; more than three quarter turns are made; or a part that will not seat after three turns is left in the nest instead of going back in the tote.
- **SOP rule broken:** Steps 2.2 and 2.3, lower the part into the nest and, if it sits proud, make up to three quarter turns, then return it to the tote if it still will not seat.
- **Coaching note:** the nest tells you which way round it is. Turn it a quarter at a time, and give up after three.

Violation: Worked on the wrong lamp colour

- **Visible cue:** the **right gripper** takes the part out of the nest or reaches over the near rail while the lamp is amber; the **left gripper** works the tote or the nest while the lamp is green;

or both arms are in motion at the same time.

- **SOP rule broken:** Steps 2.1, 3.1, 4, and 6, the **left gripper** stages on amber and the **right gripper** loads on green, one arm moving at a time.
- **Coaching note:** amber stages, green loads. Watch the lamp, not the belt.

Violation: Loaded without matching the belt

- **Visible cue:** the **right gripper** holds still over the moving belt while the part comes down; the part scrapes, skids, or tips as it meets the belt; or the gripper stops travelling while the part is still down on the belt.
- **SOP rule broken:** Step 3.3, travel along with the mark at belt speed and only then lower the part down onto the belt.
- **Coaching note:** move with the mark first, come down second. A part set on a belt you are not matching will always skid.

Violation: Released above the belt or dragged on release

- **Visible cue:** the **right gripper** opens before the part's base is down on the belt so the part drops; the gripper opens and stays put so the part is dragged or knocked as it travels out of it; or the gripper knocks the part over as it lifts away.
- **SOP rule broken:** Step 3.4, open the gripper only once the part is down and travelling, then lift straight up off the belt.
- **Coaching note:** down first, open second, straight up third. A dropped part is a misfeed you then have to pick up.

Violation: Part not landed on its mark

- **Visible cue:** the part stands before its white bar, after it, straddling two bars, or on a rail when the **right gripper** comes away.
- **SOP rule broken:** Steps 3.3 and 3.4, the part is set down over the white bar of the mark it was loaded for.
- **Coaching note:** line the base up on the bar before you come down, not after.

Violation: Wrong loaded orientation

- **Visible cue:** a part is left on the belt base up, on its side, or with its notch facing the far rail, the entry flag, or the transfer flag instead of the front.
- **SOP rule broken:** Steps 3.2 and 3.4, the part goes onto the belt base down, boss up, notch toward the front.
- **Coaching note:** the notch faces you. Check it in the gripper before you go over the rail.

Violation: Loaded a mark that already carried a part

- **Visible cue:** the **right gripper** sets a part down on the fifth mark, or on any mark that already carries a part, so two parts ride the same bar.
- **SOP rule broken:** Step 5.1, a mark that already carries a part is never loaded, and the staged part stays in the nest for the next mark.
- **Coaching note:** look at the mark before you take the part out of the nest. One bar, one part.

Violation: Misfeed not cleared to the reject bin

- **Visible cue:** the misfeed rides past the transfer flag still on the belt; it is swept, dragged, or pushed along the belt instead of lifted off; it is knocked over the far rail; or it is put down on the load shelf, back in the tote, back on the belt, or on the floor instead of in the reject bin.
- **SOP rule broken:** Steps 5.1 and 5.2, lift the misfeed straight off the belt before the transfer flag and lower it into the reject bin.
- **Coaching note:** lift it, don't shove it, and it goes in the bin every time.

Violation: Made up a lost mark

- **Visible cue:** two parts are put on one mark; a part is set down between two marks to make up the empty fifth; the **right gripper** chases the fifth mark toward the transfer flag to squeeze a part on; or a later mark is double-loaded.
- **SOP rule broken:** Step 5.3, let the fifth mark pass empty and keep the staged part in the nest for the next mark.
- **Coaching note:** a lost mark stays lost. Take the next one clean.

Violation: Takt not kept

- **Visible cue:** a pitch mark other than the fifth goes past the transfer flag empty; a mark is skipped; the **right gripper** is still in the nest or on the shelf when a green ends; or a ninth mark is loaded after the eighth.
- **SOP rule broken:** Steps 4 and 6, take every mark in turn and let no mark pass empty, and stop after the eighth.
- **Coaching note:** stage on the amber you are given, so the green is only ever a load. Eight marks and no more.

Violation: Released part not watched down the window

- **Visible cue:** the **right gripper** turns away to the nest or the tote as soon as it releases; a part tips, turns, or slides off its bar before the transfer flag and is left to go past; or a misfeed is spotted only after it has passed the flag.
- **SOP rule broken:** Steps 3.5 and 4, hold the **right gripper** clear and watch each part all the way to the transfer flag.
- **Coaching note:** watch it land and watch it go. The flag is your last chance to catch it.

Violation: Nest, tote, or bin disturbed

- **Visible cue:** the parts tote, the nest, or the reject bin is pushed out of place, tipped, lifted off the load shelf, or knocked so parts spill; or a part is left in the nest at the end of the episode.
- **SOP rule broken:** Steps 2.1 to 2.4, 5.2, and 7, the tote, the nest, and the bin stay where they stand, and the nest is empty when the episode ends.
- **Coaching note:** reach into them, never move them. They are set where the arms can reach them.

Violation: Wrong order of work

- **Visible cue:** an arm reaches in before the panel and lamp are read; a part is taken out of the nest before it is seated; a part goes over the near rail before the lamp is green; the misfeed is cleared before the staged part is in the nest; or a mark is loaded before the one before it has gone by.
- **SOP rule broken:** Steps 1 to 6, check the line, stage on amber, load on green, watch it down, clear the misfeed on the fifth, and keep the takt to the eighth.
- **Coaching note:** the order is the takt. Every step has its own lamp colour.

Violation: Required check not followed

- **Visible cue:** a check named in a step is skipped, or a check is made and the fault it finds is left uncorrected — an unseated part loaded anyway, a part left off its mark, a part left turned the wrong way, a misfeed spotted and left on the belt, or a part left in the nest.
- **SOP rule broken:** Steps 1.1 to 6, run each check and correct what it finds by the retry written in that step.
- **Coaching note:** a check is not done until what it found has been put right.

Violation: Dropped, spilled, or knocked over

- **Visible cue:** a part is dropped on the load shelf, on the belt, over the far rail, or on the floor; the tote or the bin is knocked over; or an arm knocks the entry flag, the transfer flag, or the near rail.
- **SOP rule broken:** Steps 2 to 6, nothing is dropped or knocked out of its place, and every gripper leaves the belt straight up and back out over the near rail.
- **Coaching note:** check the path and the landing place before the arm moves, and come out the way you went in.

Violation: Wrong arm used

- **Visible cue:** the **left gripper** goes over the near rail, touches the belt, or touches the reject bin; the **right gripper** reaches into the parts tote; or a part is handed from one gripper to the other instead of going through the nest.
- **SOP rule broken:** Steps 2 to 6, the **left gripper** works the tote and the nest, the **right gripper** works the nest, the belt, and the reject bin, and every part passes through the nest.

- **Coaching note:** left stages, right loads, and the nest is the only place they meet.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a part still in the nest or still in a gripper, with the misfeed not in the reject bin, with fewer or more than eight marks gone by, with a part left on a rail, with an arm short of home, or with a gripper not fully open.
- **SOP rule broken:** Step 7, confirm the line, the marks, the belt, the shelf, the drops, and the base, then return both arms home with grippers open and stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Camera cannot read the pitch marks, the takt lamp, or the nest**, so whether a part landed on its mark, which lamp colour a move was made on, or whether a part was seated cannot be judged.
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Base fault:** a brake or lock that will not hold, or a base that rolls, creeps, or turns with the lock set.
- **Line stopped by itself** mid-episode, or the belt jammed, slipped, or changed speed on its own.
- **Belt too fast for the station:** a set speed at which a pitch mark crosses the load window before one part can be taken from the nest and set down.
- **Faulty takt lamp:** one that does not change with the marks, that changes out of step with them, or that is unreadable from the front.
- **Faulty panel:** a panel that does not read the line running, or that is unreadable from the front.
- **Pitch marks unusable:** bars worn faint, unevenly spaced, or missing from stretches of the belt.
- **Faulty nest:** a nest whose rib is worn down so a part sits flat in it the wrong way round, or whose pocket holds a part so tight it cannot be lifted straight out.
- **Defective part:** one that arrives with a broken boss, a base that will not stand flat, or a notch too worn to read.
- **Loose flag:** an entry flag or transfer flag that falls, slides along the rail, or is missing, so the load window has no end.

- **A place turns out to sit outside its arm's comfortable reach** with the base correctly parked, so the bottom of the tote, the nest, the reject bin, or a mark at either end of the load window cannot be reached without extending or folding the arm.

Annotation subtasks (from SOP)

1. Read the panel and the takt lamp and check the line is running
2. Take one part out of the parts tote
3. Lower the part into the nest
4. Turn a part a quarter turn in the nest
5. Put an unseated part back in the parts tote
6. Take the seated part out of the nest
7. Carry the part in over the near rail
8. Travel along with the pitch mark at belt speed
9. Set the part down on the pitch mark
10. Release the part and lift clear of the belt
11. Watch a loaded part down to the transfer flag
12. Wait on amber with both arms clear
13. Wait on green with the staged part left in the nest
14. Take the misfeed off the belt
15. Drop the misfeed in the reject bin
16. Let a pitch mark pass empty
17. Load one pitch mark (repeat for marks 2 to 4)
18. Load one pitch mark (repeat for marks 6 to 8)
19. Draw both arms clear after the eighth mark
20. Return both arms home and end the episode